

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Inference from sample statistics and margin of error

01

10 answers · Rationale-rich · Teach from doc

Q1**A**

A. It is plausible that the proportion is between 0.45 and 0.53.

Choice A is correct. It's given that the estimate for the proportion of the population that has the characteristic is 0.49 with an associated margin of error of 0.04. Subtracting the margin of error from the estimate and adding the margin of error to the estimate gives an interval of plausible values for the true proportion of the population that has the characteristic. Therefore, it's plausible that the proportion of the population that has this characteristic is between 0.45 and 0.53.

Choice B is incorrect. A value less than 0.45 is outside the interval of plausible values for the proportion of the population that has the characteristic.

Choice C is incorrect. The value 0.49 is an estimate for the proportion based on this sample. However, since the margin of error for this estimate is known, the most appropriate conclusion is not that the proportion is exactly one value but instead lies in an interval of plausible values.

Choice D is incorrect. A value greater than 0.53 is outside the interval of plausible values for the proportion of the population that has the characteristic.

Q2

A

- A. It is plausible that the actual mean value of the variable for the population is between 19.5 and 21.5.
-

Choice A is correct. It's given that based on a random sample from a population, the estimated mean value for a certain variable for the population is 20.5, with an associated margin of error of 1. This means that it is plausible that the actual mean value of the variable for the population is between $20.5 - 1$ and $20.5 + 1$. Therefore, the most appropriate conclusion is that it is plausible that the actual mean value of the variable for the population is between 19.5 and 21.5.

Choice B is incorrect. The estimated mean value and associated margin of error describe only plausible values, not all the possible values, for the actual mean value of the variable, so this is not an appropriate conclusion.

Choice C is incorrect. The estimated mean value and associated margin of error describe only plausible values for the actual mean value of the variable, not all the possible values of the variable, so this is not an appropriate conclusion.

Choice D is incorrect. Since 20.5 is the estimated mean value of the variable based on a random sample, the actual mean value of the variable may not be exactly 20.5. Therefore, this is not an appropriate conclusion.

Q3

D

- D. Approximately 30% of all largemouth bass in the pond weigh more than 2 pounds.
-

Choice D is correct. The sample of 150 largemouth bass was selected at random from all the largemouth bass in the pond, and since 30% of the fish in the sample weighed more than 2 pounds, it can be concluded that approximately 30% of all largemouth bass in the pond weigh more than 2 pounds.

Choices A, B, and C are incorrect. Since the sample contained 150 largemouth bass, of which 30% weighed more than 2 pounds, this result can be generalized only to largemouth bass in the pond, not to all fish in the pond.

Q4

D

- D. The best estimate for the number of customers who purchased fruit last Thursday is 375.
-

Choice D is correct. It's given that the manager took a random selection of the receipts of 80 customers from a total of 1,500. It's also given that of those 80 receipts, 20 showed that the customer had purchased fruit. This means that an appropriate estimate of the fraction of customers who purchased fruit is $\frac{20}{80}$, or $\frac{1}{4}$. Multiplying this fraction by the total number of customers yields $\left(\frac{1}{4}\right)(1,500) = 375$. Therefore, the best estimate for the number of customers who purchased fruit is 375.

Choices A and B are incorrect because an exact number of customers can't be known from taking a random selection. Additionally, choice A may also be the result of a calculation error. Choice C is incorrect and may result from a calculation error.

Q5**B**

B. It is plausible that between 8.38% and 15.62% of the socks are defective.

Choice B is correct. It's given that, based on the sample, an estimate of 12% of all socks produced by the company in a certain week are defective, with an associated margin of error of 3.62%. This estimate, plus or minus the margin of error, gives an interval of plausible values for the actual percent of all socks produced by the company that week that are defective.

Subtracting 3.62% from 12% yields 8.38%. Adding 3.62% to 12% yields 15.62%. Therefore, it is plausible that between 8.38% and 15.62% of all socks produced by the company are defective.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. 12% is the estimated percent of defective socks based on the sample. However, since the margin of error for this estimate is known, the most appropriate conclusion is not that the percent of defective socks is exactly 12% but instead that it lies in an interval of plausible percents.

Choice D is incorrect and may result from conceptual errors.

Q6**3**

The correct answer is 3,540. According to the table, of 400 voters randomly sampled, the total number of men and women who plan to vote for Candidate A is $202 + 34 = 236$. The best estimate of the total number of voters in the town who plan to vote for Candidate A is the fraction of voters in the sample who plan to vote for Candidate A, $\frac{236}{400}$, multiplied by the total

voter population of 6000. Therefore, the answer is $\left(\frac{236}{400}\right)(6,000) = 3,540$.

Q7

A

- A. Plausible values for the average weight of all boxes of oranges filled by the company are between 22.91 pounds and 23.29 pounds.
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Choice A is correct. It's given that the estimate for the average weight of all boxes of oranges filled by the company in an 8-hour period is 23.1 pounds, with an associated margin of error of 0.19 pounds. It follows that plausible values for this average weight are between $23.1 - 0.19$ pounds and $23.1 + 0.19$ pounds. Therefore, plausible values for the average weight of all boxes of oranges filled by the company are between 22.91 pounds and 23.29 pounds.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

B

- B. Between 28.59% and 35.41% of all adults who own televisions use their televisions to watch nature shows.
-

Choice B is correct. It's given that based on a sample selected at random, it's estimated that 32% of all adults who own televisions use their televisions to watch nature shows, with an associated margin of error of 3.41%. Subtracting the margin of error from the estimate and adding the margin of error to the estimate gives an interval of plausible values for the true percentage of adults who own televisions who use their televisions to watch nature shows. This means it's plausible that between $32\% - 3.41\%$, or 28.59%, and $32\% + 3.41\%$, or 35.41%, of all adults who own televisions use their televisions to watch nature shows. Therefore, of the given choices, the most plausible conclusion is that between 28.59% and 35.41% of all adults who own televisions use their televisions to watch nature shows.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. To make a plausible conclusion about all adults who own televisions, the sample must be selected at random from all adults who own televisions, not just those who use their televisions to watch nature shows.

Choice D is incorrect. Since the sample was selected at random from all adults who own televisions, a plausible conclusion can be made about all adults who own televisions.

Q9**C****C.** 16,750

Choice C is correct. It's given that an estimated 35% of people in the population support the legislation, with an associated margin of error of 3%. Subtracting and adding the margin of error from the estimate gives an interval of plausible values for the true percentage of people in the population who support the legislation. Therefore, it's plausible that between 32% and 38% of people in this population support the legislation. The corresponding numbers of people represented by these percentages in the population can be calculated by multiplying the total population, 50,000, by 0.32 and by 0.38, which gives $50,000 \cdot 0.32 = 16,000$ and $50,000 \cdot 0.38 = 19,000$, respectively. It follows that any value in the interval 16,000 to 19,000 is a plausible value for the total number of people in the population who support the proposed legislation. Of the choices given, only 16,750 is in this interval.

Choice A is incorrect. This is the number of people in the sample, rather than in the population, who support the legislation.

Choice B is incorrect. This is the number of people in the sample who do not support the legislation.

Choice D is incorrect. This is a plausible value for the total number of people in the population who do not support the proposed legislation.

Q10**D****D.** It is plausible that the mean distance hiked for all visitors is between 4 and 5 miles.

Choice D is correct. The given estimated mean has an associated margin of error because from sample data, the population mean can't be determined precisely. Rather, from the sample mean, an interval can be determined within which it's plausible that the population's mean is likely to lie. Since the estimated mean is 4.5 miles with an associated margin of error of 0.5 miles, it follows that between $4.5 - 0.5$ miles and $4.5 + 0.5$ miles, or between 4 and 5 miles, is plausibly the mean distance hiked for all visitors.

Choices A, B, and C are incorrect. Based on the estimated mean, no determination can be made about the number of miles hiked for all visitors to the park.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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